

Full Stack Development with MERN

Project Documentation

1. Introduction

- **Project Title:** HouseHunt - Finding Your Perfect Rental Home
- **Team Members:**

Name	Role	Responsibilities
Somala Kanth Mani Sai	Full Stack MERN Developer	Designed system architecture, developed frontend (React), implemented backend APIs (Node.js & Express), designed MongoDB schemas, handled authentication, testing, and documentation

2. Project Overview

- **Purpose:**

HouseHunt is a full-stack MERN web application developed to simplify the process of renting properties. It connects property owners and renters on a centralized digital platform where:

 - Owners can add and manage properties
 - Renters can browse and book properties
 - Admin can monitor system activities

The goal of this project is to provide a secure, scalable, and user-friendly rental management system.
- **Key Features:**
 - ✓ User Registration & Login (JWT Authentication)
 - ✓ Role-based access (Admin, Owner, Renter)
 - ✓ Add / Update / Delete Property
 - ✓ View Available Properties
 - ✓ Property Booking System
 - ✓ Admin Dashboard
 - ✓ Secure API with Middleware Protection
 - ✓ MongoDB Database Integration

3. Architecture

- **Frontend:**
 - Built using **React.js**
 - Routing handled using **React Router DOM**
 - Axios used for API communication
 - Role-based page rendering

- Token stored in localStorage
- Modular folder structure:
 - common (Login, Register, Home)
 - user → owner / renter
 - admin dashboard
- Frontend communicates with backend using REST APIs.

- **Backend:**

- Built using Node.js and Express.js
- RESTful API architecture
- MVC Pattern:
 - Models (MongoDB Schemas)
 - Routes
 - Controllers
- Middleware (Authentication & Authorization)
- JWT-based authentication
- Protected routes using middleware

- **Database (MongoDB):**

MongoDB is used as the NoSQL database.

Main Collections:

1. Users Collection

```
{
  name: String,
  email: String,
  password: String,
  role: "admin" | "owner" | "renter"
}
```

2. Properties Collection

```
{
  type: String,
  address: String,
  amount: Number,
  contact: String,
  ownerId: ObjectId,
  isAvailable: Boolean
}
```

3. Bookings Collection

```
{
  propertyId: ObjectId,
  renterId: ObjectId,
  bookingDate: Date
}
```

4. Setup Instructions

- **Prerequisites**

- Node.js (v18+)
- MongoDB installed locally or MongoDB Atlas
- npm
- Git

▪ **Installation Steps**

Step 1: Clone the Repository

```
git clone <repository-url>
```

```
cd househunt
```

Step 2: Setup Backend

```
cd backend
```

```
npm install
```

Create .env file:

```
PORT=5000
```

```
MONGO_URI=mongodb://127.0.0.1:27017/househunt
```

```
JWT_SECRET=your_secret_key
```

Start backend: npm start

Step 3: Setup Frontend

```
cd frontend
```

```
npm install
```

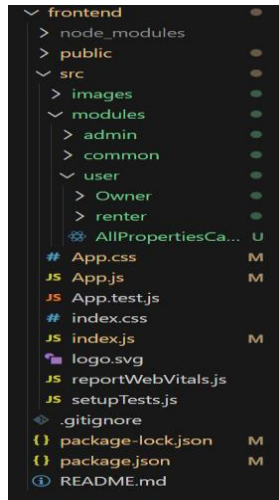
```
npm start
```

Frontend runs on: http://localhost:3000

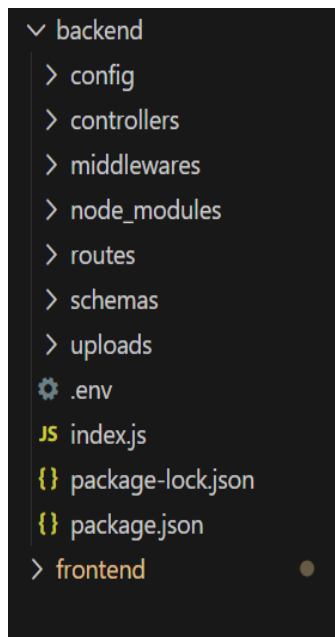
Backend runs on: <http://localhost:5000>

5. Folder Structure

Client (React Frontend)



Server (Node Backend)



6. Running the Application

Backend

```
cd backend
```

```
npm start
```

Frontend

```
cd frontend
```

```
npm start
```

7. API Documentation

➤ User APIs

Method	Endpoint	Description
POST	/api/users/register	Register new user
POST	/api/users/login	Login user

➤ Owner APIs

Method	Endpoint	Description
POST	/api/owner/property	Add property
GET	/api/owner/property	Get owner properties
PUT	/api/owner/property/:id	Update property
DELETE	/api/owner/property/:id	Delete property

➤ Renter APIs

Method	Endpoint	Description
GET	/api/renter/property	View all available properties
POST	/api/renter/book/:id	Book property

8. Authentication

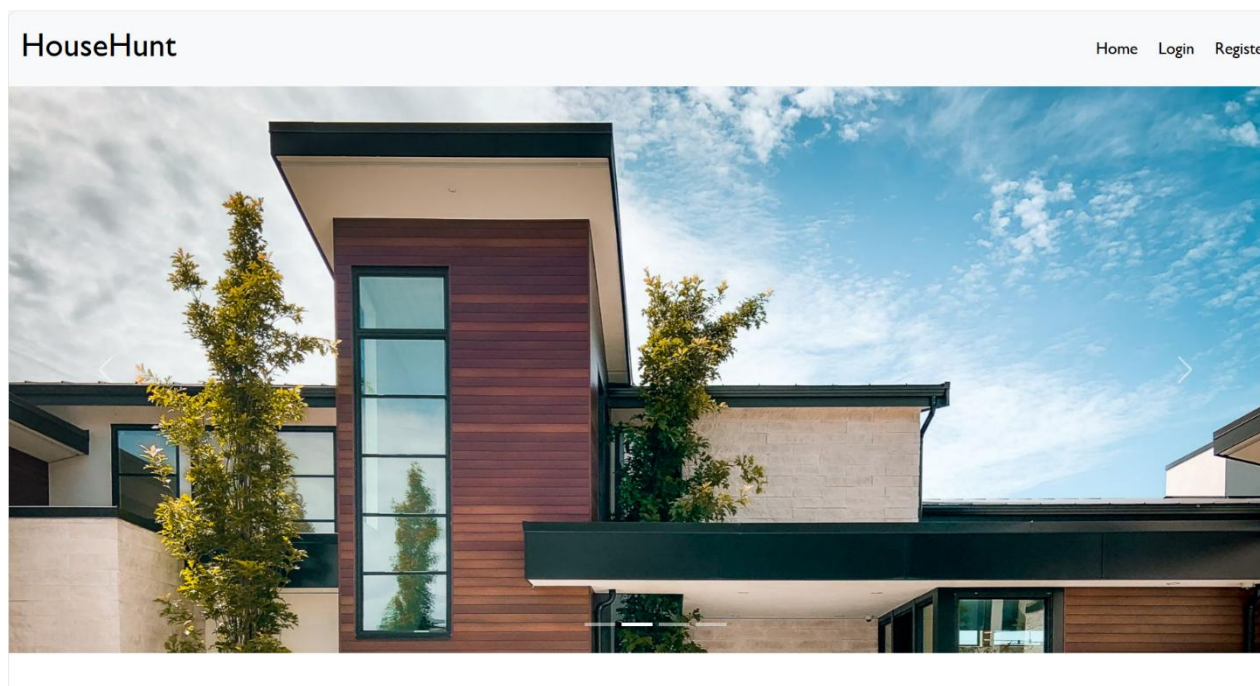
Authentication is handled using JWT (JSON Web Token).

Process:

1. User logs in
2. Backend generates JWT token
3. Token stored in localStorage
4. Token sent in Authorization header
5. Middleware verifies token before accessing protected routes

9. User Interface

➤ Home Page



➤ Login Page

HouseHunt

HomeLoginRegister

Sign In

Email Address

Password

SIGN UP

forgot password? [Click here](#) Have an account? [Sign Up](#)

© 2026 Copyright: HouseHunt

➤ Register Page

HouseHunt

HomeLoginRegister

Sign up

Renter Full Name/Owner Name

Email Address

Password

User Type

SIGN UP

Have an account? [Sign In](#)

© 2026 Copyright: HouseHunt

➤ Owner Dashboard/Add Property Page

HouseHunt

Hi Sriram Log Out

ADD PROPERTYALL PROPERTIESALL BOOKINGS

Property type

Residential

Property Ad type

Rent

Property Full Address

Address

Property Images

Choose FilesNo file chosen

Owner Contact No.

contact number

Property Amt.

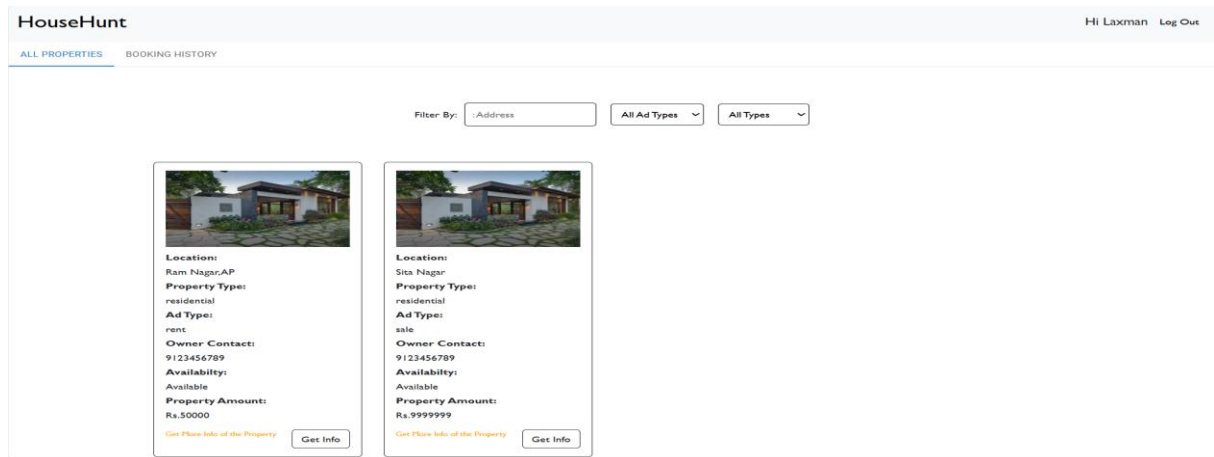
0

Additional details for the Property

Submit form

© 2026 Copyright: HouseHunt

➤ Renter Dashboard



10. Testing

Testing Strategy:

- Manual Testing
- Unit testing of APIs
- UAT Testing
- JWT Validation Testing
- Role-based Access Testing

Tools Used:

- Postman (API testing)
- Browser Developer Tools
- MongoDB Compass

11. Demo

Project Demo Link:

https://drive.google.com/file/d/1qkn8_kludKuzHpqVZkd5G0gvJtujunW6/view?usp=drivesdk

12. Known Issues

- No email verification implemented.
- UI can be further enhanced using advanced styling.
- No payment gateway integration.
- Limited filtering options for properties.

13. Future Enhancements

- Add payment gateway integration
- Add property images upload
- Implement email verification
- Add property search & filters
- Add real-time notifications
- Implement chat between renter and owner